Statistical analysis of PIFs inversions

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Goal

Hypothesis: Shorter distance and more adjacent legs $\Rightarrow$ more inversions.

Goal: to support validity of hypothesis with data.

PIFs inversion measure

- Analyze inversion corrected PIFs before modified by users
- Current inversion algorithm produce long sequences of same values when PIFs inverted
- Longer sequence $\Rightarrow$ more inversions
- One year data analyzed
- Exclude legs with length $>=$ 3000 km

Measure *pi*: number of distinct PIFs.

Greater *pi* $\Rightarrow$ less inversions and vice versa.

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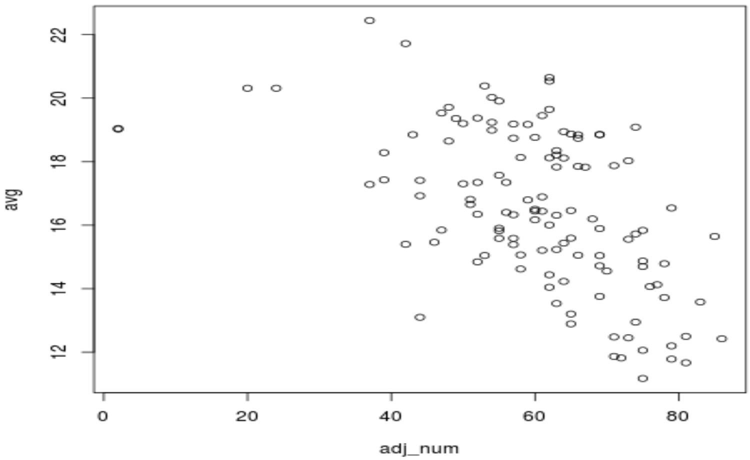
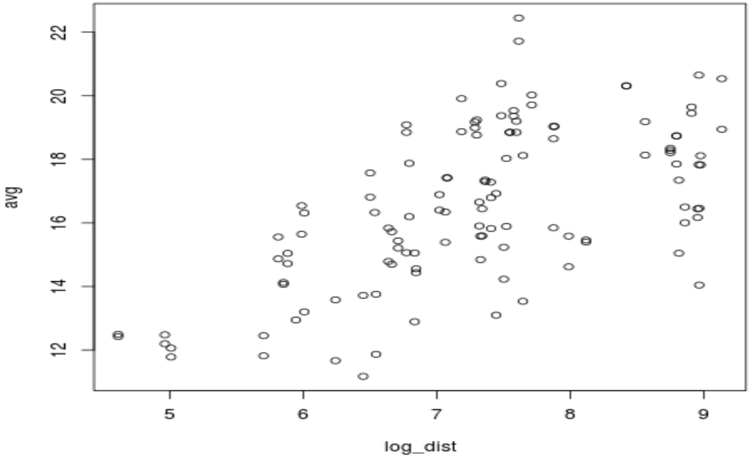
Features

- avg_i — sample mean of pi for leg i
- $adjnum_i$ is number of adjacent legs to leg i
- $dist_i$ is distance to fly over leg i

pi vs adjnum



pi vs log(dist)



Regression

Model:

$$pi = a_0 + a_1 * log(dist) + a_2 * adjnum$$

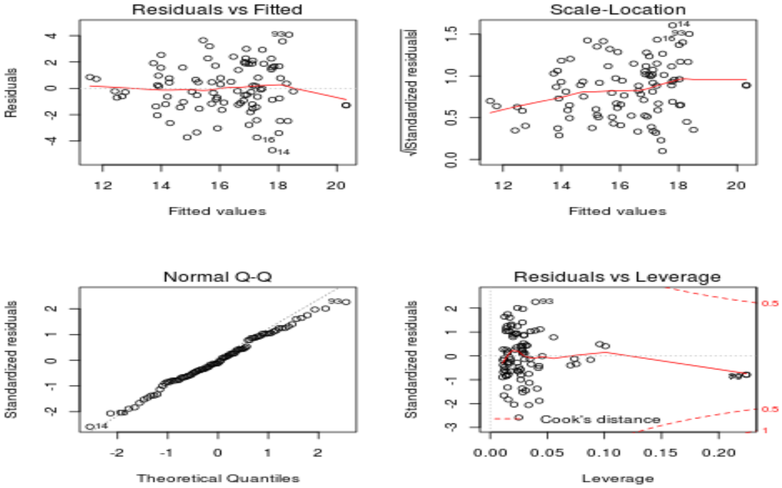
Fit:

$$pi = 8.2 + 1.5 * log(dist) - 0.05 * adjnum$$

Quality:

$$R^2 = 0.48$$

Residuals


Residuals vs Fitted
Residuals
Fitted values
93
16
14
Scale-Location
√|Standardized residuals|
Fitted values
Normal Q-Q
Standardized residuals
Theoretical Quantiles
Residuals vs Leverage
Standardized residuals
Leverage
Cook's distance
0.5
1

Conclusions

- Longer leg $\Rightarrow$ less inversions
- Greater number of adjacent legs $\Rightarrow$ more inversions